

## E.2 Effect of Concept Conditioning by Model

Table A.3 details the impact of concept conditioning for each model, representation, and category. These numbers underlie the aggregated trends: concept conditioning has small, model- and category-dependent effects, with Action Programs often showing slightly more stable behavior than Action Descriptions in the Human-designed regime.

Concept conditioning yields differential benefits: Action Program representations benefit more (+3.2% avg) than Action Description (+1.3%). Abstract Shape problems show largest gains (+5.9% for AP), Free-form problems minimal gain (+0.9%).

## E.3 Effect of Symbolic Representation Format

Natural-language descriptions provide strong inductive bias for the easier FF and BD regimes but fail to generalize to Human-designed abstractions. In contrast, action programs yield smaller but consistent gains, motivating their use as the primary concept representation. Overall, format choice has a modest impact relative to the much larger gap between symbolic and visual-only input.

## E.4 Effect of Model Capacity

We observe that the effect of symbolic input and concept conditioning varies across models (Table A.4). While some smaller and mid-sized models show clear gains over their visual baselines, others see little change or even slight decreases, and the larger models in our analysis tend to exhibit modest but more consistently positive improvements. This pattern is compatible with the view that structured input can act as an inductive bias that reduces representational burden, but its benefits are not uniform across capacity tiers.

Table A.4: Mean accuracy improvement (%) over baseline across all regimes, stratified by model. The effect of symbolic input and concept conditioning is heterogeneous: larger models show slightly more consistent positive averages, while smaller and instruction-sensitive models exhibit mixed gains.

| Model                  | AD   | AP   | AD + C | AP + C |
|------------------------|------|------|--------|--------|
| DeepSeek-R1:14b        | -0.7 | -0.3 | +0.0   | -0.5   |
| DeepSeek-R1:32b        | +3.6 | +3.1 | +1.7   | +2.8   |
| DeepSeek-R1:8b         | +8.0 | +3.5 | +4.8   | +3.4   |
| Gemini-3-flash-preview | +0.3 | +1.3 | —      | —      |
| Gemma3:12b             | -0.4 | +0.7 | -0.6   | +0.0   |
| Gemma3:27b             | -2.9 | +0.8 | -0.4   | +1.1   |
| Magistral              | +0.0 | -2.5 | +1.2   | -0.5   |
| Phi4                   | -4.1 | -1.8 | -3.5   | -1.1   |
| Phi4-Reasoning         | -1.7 | +2.5 | +1.1   | +2.2   |
| Qwen2.5:32b            | +2.1 | +1.6 | +0.5   | +2.9   |
| Qwen2.5vl:32b          | +4.5 | +4.4 | +0.4   | +4.6   |
| Qwen3:30b              | -0.1 | +2.3 | +2.0   | +0.3   |
| Qwen3:32b              | +2.4 | -1.3 | +2.2   | +2.7   |

To analyze how symbolic input interacts with model capacity, we partition models into *smaller* and *larger* capacity tiers. This grouping is based on a combination of parameter scale, reasoning-oriented pretraining, and baseline performance on Bongard-LOGO. The grouping is fixed across all experiments and comparisons.

Table A.5: Model grouping used for capacity-based analysis. Models are grouped by approximate parameter scale and reasoning capacity.

| Capacity Tier | Models                                                                         |
|---------------|--------------------------------------------------------------------------------|
| Smaller       | Phi-4-Reasoning, Phi-4, Magistral, Gemma3:12B, DeepSeek-R1:7B, DeepSeek-R1:14B |
| Larger        | Qwen3:32B, Qwen3:30B, Gemma3:27B, DeepSeek-R1:32B, Qwen2.5vl:32B, Qwen2.5:32B  |

To further analyze the interaction between model capacity and symbolic input, we group models into smaller and larger tiers and average accuracy improvements within each group. This grouping is based on a combination of parameter scale as shown in Table A.5. Accuracy is presented in Table A.6. Larger models in this dataset show slightly higher mean gains from symbolic input and concept conditioning than the smaller group, although variance across individual models remains substantial. These exploratory results suggest that the benefits of symbolic input depend on both capacity and pretraining, rather than uniformly favoring lower-capacity models.

Table A.6: Mean accuracy improvement (%) over baseline, averaged across model size groups. In this dataset, larger models show slightly higher mean gains from symbolic input and concept conditioning than the smaller group.

| Model Group    | AD         | AP         | AD + Concept | AP + Concept |
|----------------|------------|------------|--------------|--------------|
| Smaller Models | +0.2 ± 3.7 | +0.4 ± 2.2 | +0.5 ± 2.5   | +0.6 ± 1.6   |
| Larger Models  | +1.6 ± 2.5 | +1.8 ± 1.8 | +1.1 ± 0.9   | +2.4 ± 1.4   |

### E.5 Effect of Grounded C–G with Query Image Anchors

The Grounded C–G condition augments symbolic input with a single visual anchor: the rendered query image. This condition is only applicable to models that support image inputs. We therefore restrict this analysis to VLM-capable models: Gemma3 12B, Gemma3 27B, LLaVA-Llama3, and Qwen2.5-VL 32B.

In all cases, the model receives (i) the action programs or action descriptions for the six positive and six negative examples, and (ii) the corresponding symbolic representation plus rendered image for the query. The task and evaluation protocol are identical to the main experiments (Section 5).

Table A.7 reports per-model performance across Bongard categories for the two symbolic formats, and Table A.8 summarizes mean accuracy across the four VLMs, aggregating Human-designed Combined and Novel (HD Comb, HD Novel) into a single HD score. Note that, unlike Table 1, which averages over 12 models, the statistics here are computed over this smaller matched VLM subset and should be compared only within that subset.

For ease of comparison with the main results (Table 1), we also report mean accuracy and standard deviation across the four VLMs, averaging HD Comb and HD Novel into a single Human-designed score:

Overall, the grounded setting yields only modest changes relative to the corresponding symbolic-only baselines for this matched VLM subset, and Action Programs retain a small advantage on Free-form and Human-designed categories. This reinforces our main conclusion that symbolic structure, rather than limited visual anchoring, is the primary source of gains on Bongard-LOGO.

### E.6 Randomization Experiments

To probe whether high C–G performance reflects genuine rule learning over symbolic structure rather than exploitation of superficial token statistics, we conduct two perturbation experiments in the Action Program setting:

- **Categories Shuffle:** for each Bongard-LOGO problem, we randomly permute the assignment of the 12 support examples to the positive and negative sets while leaving the query program, query image, and gold query label unchanged. This preserves marginal token distributions within a problem but destroys the coherent support-set partition that defines the concept.
- **Test Sequence Shuffle:** for each problem, we randomly shuffle the order of drawing actions in the *query* program at inference time, while leaving the support examples, query image, and gold query label unchanged. This maintains the multiset of action tokens but disrupts the compositional structure of the query.

Both perturbations use the same prompts and decoding settings as the main Action Program condition (Section 4). We report them only on Basic (BD) and Free-form (FF), since the lower unperturbed HD baseline would make the controls less diagnostic there. Table A.9 reports per-model accuracies on BD and FF, and Table A.10 summarizes mean accuracy across the nine models included in this analysis.

Table A.7: Grounded C–G performance for VLM-capable models. Each entry is accuracy (%) on Bongard-LOGO categories when symbolic input (Action Description or Action Program) is augmented with the rendered query image. BD = Basic, FF = Free-form, HD Comb/HD Novel = Human-designed shape subsets.

| Model         | Experiment  | BD    | FF   | HD Comb | HD Novel |
|---------------|-------------|-------|------|---------|----------|
| Gemma3:12B    | Action Desc | 55.0  | 56.2 | 56.8    | 56.0     |
|               | Action Prog | 56.31 | 58.8 | 57.2    | 60.12    |
| Gemma3:27B    | Action Desc | 56.2  | 69.2 | 57.6    | 61.0     |
|               | Action Prog | 59.0  | 70.4 | 65.0    | 63.4     |
| LLaVA-Llama3  | Action Desc | 49.6  | 50.1 | 50.0    | 50.1     |
|               | Action Prog | 50.2  | 49.0 | 49.2    | 50.2     |
| Qwen2.5VL:32B | Action Desc | 60.4  | 69.2 | 57.8    | 57.4     |
|               | Action Prog | 55.8  | 71.2 | 59.6    | 61.0     |

Table A.8: Grounded C–G: mean accuracy (%)  $\pm$  standard deviation across VLM-capable models (N=4). HD is the average of HD Comb and HD Novel. These numbers are not directly comparable to the 12-model pooled rows in Table 1.

| Experiment         | BD             | FF              | HD             |
|--------------------|----------------|-----------------|----------------|
| Action Description | 55.3 $\pm$ 4.5 | 61.2 $\pm$ 9.6  | 55.8 $\pm$ 4.0 |
| Action Program     | 55.3 $\pm$ 3.7 | 62.4 $\pm$ 10.6 | 58.2 $\pm$ 6.1 |
| Baseline           | 57.1 $\pm$ 5.5 | 62.6 $\pm$ 11.2 | 57.5 $\pm$ 5.4 |

Overall, Categories Shuffle and Test Sequence Shuffle both reduce accuracy relative to the unperturbed Action Program setting, but they need not drive performance exactly to 50% because the shuffled support sets can still instantiate alternative rules that partially align with the unchanged query. The relevant signal is the relative degradation: Categories Shuffle primarily damages coherent support-set reasoning, whereas Test Sequence Shuffle particularly harms Free-form performance by disrupting the ordered structure of the query. These effects support the interpretation that models rely on the relational and compositional structure of the symbolic programs, especially for Free-form tasks, rather than on simple surface-level token matching.

## E.7 Class Asymmetry in Rule Induction

Beyond overall accuracy, we analyze how models treat positive vs. negative examples under the Action Program (AP) interface. Aggregating across models, we find a pronounced asymmetry: on Basic (BD) problems, accuracy on positive examples is 91.2% compared to 53.4% on negatives (a +37.8 point gap), and on Free-form (FF) problems, positives reach 98.5% vs. 61.4% for negatives (+37.1 points). A chi-square test of correctness  $\times$  class yields  $\chi^2 \approx 1156.3$  for BD and  $\chi^2 \approx 1504.2$  for FF ( $p \ll 0.001$  in both cases), indicating that correctness is strongly dependent on whether an item belongs to the positive or negative set. Consistent with this, the pooled predicted labels are skewed toward “positive” (e.g., 4,799 predicted positives vs. 2,201 negatives on FF), suggesting that models default to classifying borderline or ambiguous cases as positive rather than negative.

## F Limitations

### F.1 Synthetic Benchmark Limitations

While Bongard-LOGO provides a controlled environment for hypothesis testing, it is fundamentally limited:

- Simple geometric shapes with programmatic rules (not learned from data)
- Absence of real-world noise, occlusion, variation, and clutter
- Binary classification rather than open-ended reasoning